HOUSE FORMULA / V2.1 · BULK METHOD

NYC-Style Bagels

Straight dough · short ~3h bulk → shape → **emphasized pan proof** → cold hold · boiled & baked

YIELD	FLOUR	HYDRATION	METHOD	TOTAL DOUGH
24 @ 150g	**Sir Lancelot 14.2%**	**60%**	**Straight + bulk**	**~3,630g**

STRAIGHT DOUGH

No poolish — mix it all at once

Sir Lancelot	**2,200 g**
Water (60% — cold if warm kitchen)	**1,320 g**
Salt (2%)	**44 g**
Honey (~2.5%)	**55 g**
ADY (0.6%)	**13.2 g**

The flour's malted barley supplies the enzyme load. **0.6% ADY targets a ~4.5h total room ferment** (~3h bulk + ~1.5h pan proof) — ~6× the pizza 0.2% because this runs ~5× faster. Derived against the TXCraig1 yeast model at a granite-cooled ~66°F effective bulk temp. Start here; if the bulk hits 50% well before 3h, drop toward 0.4% — if it drags, the granite's colder than figured, nudge up or extend.

MIX

Two batches — incorporation only, then fold

Mix straight, then divide by weight for the stretch-and-fold. Each batch makes 12 bagels.

PER MIXING BATCH ×2

Flour	**1,100 g**
Water	**660 g**
Salt	**22 g**
Honey	**27.5 g**
ADY	**6.6 g**

Mixer = incorporation only. At 60% with 14.2% flour the hook can't reach windowpane — gluten's too strong to fully develop in the machine. Run low just until combined, no dry bits, then stop.

Develop by stretch-and-fold: 2–3 S&F sets ~30 min apart through the first ~1.5h of bulk; the rests autolyse, the folds build the network. **Windowpane is a post-fold check.** Low-friction mixing in a 70°F room lands the dough near room temp (~68–70°F) — already factored into the yeast and bulk timing.

~1,815g dough per batch → 12 bagels @ 150g.

> **The consistency lever:** shaping from a **bulk-fermented mass** — not pre-divided pieces — is why pro bagels come out even without a scale. A short bulk organizes the gluten across the whole dough so each piece ropes uniformly. This is the structural reason for the fork.

TEMP TARGETS

this stage — mix & dough

Room (dough rests here)	**70°F**
Dough after mix	**~68–70°F**

SHAPE · PROOF · BOIL · BAKE

Execution

ROOM (BULK+PROOF)	BULK PULL	BOIL BATH	BAKE
70°F	**~50% rise**	**212°F roll**	**450°F**

BULK · SHAPE · PROOF · HOLD

1. **Short bulk, ~3 h:** mix in the morning, bulk on the counter **to ~50% rise**. Run **2–3 S&F sets, 30 min apart**, through the first ~1.5 h; rest after. The bulk's job is to **organize the gluten**, not to do the main proof — pull by rise, not clock.
2. **Divide from the bulk** to 150g (cut from the organized mass — the consistency move). Pre-round, bench rest **10–15 min**, then **rope & weld TIGHT**: fuse the seam on un-floured counter, small ~1.5″ hole = puffs up, not out.
3. **Pan proof — THIS is the main proof, ~1–2 h:** rise the shaped bagels on half-sheets at 70°F **to final height**. This is the rise that makes the bagel; weight it here, not in the bulk. Float-test when they look ready.
4. **Fridge = HOLD, not proof.** Bagels go in **already proofed**; expect ~no rise (this fridge doesn't proof). Cold-hold for flavor + scheduling, then boil straight from cold (brief temper for the chill).

Granite note: cold counter is a heat-sink — it slows the bulk (good, keeps it partial) but unevenly. Bulk in a **bowl** for even temp if you want; the pan proof runs warmer (pan buffers the counter) which is why it finishes the rise.

BOIL & BAKE

1. **Float test** straight from cold: surfaces in ~10s → go. Sinks → temper at room temp, retest. **This gates everything.**
2. **Boil bath:** per ~7 L water add **45 g baking soda** + **40 g honey**. Gentle rolling boil, never violent.
3. **Boil 30 sec / side.** Shorter boil = thinner, more tender crust. Seed wet, straight out of the water (everything blend).
4. **Bake 450°F** on a **half-sheet pan** (parchment + cornmeal), **8 per pan**. Steam the first few min. **Steel goes one rack BELOW the pan with an air gap** — radiant mass, NOT touching the pan (pan-on-steel dries the heel). **~18 min total: flip the bagels on the pan at the halfway point** — gives the heel relief from bottom heat and evens both faces. Judge by set + even color.
5. **Rest 20–30 min** before cutting. Hot crumb reads false-gummy.

BROWNING MECHANISM

Solved — no malt syrup required

Honey deposits a sugar film → **Maillard** fuel; baking soda drives the surface **alkaline**, accelerating it (pretzel chemistry). Together they cover the color the absent malt would give. Deeper mahogany: bake the soda 300°F/1h → sodium carbonate, then use in the bath.

DIAGNOSTICS

Dull, soft, over-sprung	Under-boiled — skin never set. Add boil time.
Wrinkled, tough, squat, gummy ring under skin	Over-boiled — skin set too hard, strangled spring. Cut boil time.
Skin right, interior gummy	Under-*baked*. Add 2–3 min; leave boil alone.
Tough bite, tight crumb	Under-proofed. Extend the bulk or the shaped room-temp rise; the fridge won't proof it. Float-test before boiling.
Tough, dried heel; soft top	Pan sitting directly on the steel — conduction cooks the heel all bake. Put an **air gap** (steel a rack below), and **flip at halfway** to give the heel relief.
Tough crumb, worse when toasted	Moisture-starved — **60% is the tenderness floor**.